

DIGITAL LOGIC DESIGN

م	الدروس والمواضيع بشكل مرتب		نقاط الدروس مفصلة	مصادر تعلم
1.	Boolean Algebra	Introduction to Logic Design	➤ De Morgan's Theorem ➤ Theorems of Boolean Algebra ➤ Using Truth tables for proving theorems	• Link 1 • Link 2 • Link 3
2.	Basic Logic Gates		➤ Logical Equivalence ➤ Logic Functions ➤ All gates and truth table	• Link 1 • Link 2 • Link 3
3.	Design logic and truth and function		➤ Truth table to function & gates ➤ Function to truth table & gates ➤ Gates to truth table & function ➤ waveforms Gates	• Link 1 • Link 2 • Link 3 • Link 4
4.	Pos and sop		➤ Sum-of-Products ➤ Product-of-Sums ➤ Minterms and maxterms	• Link 1 • Link 2 • Link 3
5.	Karnaugh Maps - K-Maps		➤ Karnaugh Maps for 2&3&4 Variables ➤ Karnaugh Maps with Don't Care Conditions	• Link 1 • Link 2
6.	Don't Care Conditions	Combinational Circuits	Don't care sop and pos in Karnaugh	• Link 1 • Link 2
7.	Analysis Combinational		➤ Analysis for half and full adders	• Link 1 • Link 2
8.	Adders: Half Adder, Full Adder		➤ Half adder : (T. table and F. and G) ➤ Full adder : (T. table and F. and G)	• Link 1 • Link 2
9.	Subtractors		➤ Half sub : (T. table and F. and G) ➤ Full sub : (T. table and F. and G)	• Link 1 • Link 2
10.	Encoders & Decoders		➤ Decoder With Enabel & truth table ➤ Decoder 2*4 and 3*8 and 4*16 ➤ Decoder block diagram & Logic Diagram ➤ Encoder with Enabel & truth table ➤ Encoder 4*2 and 8*3 and 16*4 ➤ Encoder block diagram & Logic Diagram	• Link 1 • Link 2 • Link 3 • Link 4
11.	Multiplexers & Demultiplexers		➤ Multiplexer truth table and block diagram ➤ Multiplexer 2*1 and 4*1 and 8*1 ➤ Bolean Function Implementation ➤ DeMultiplexer truth table and block diagram ➤ DeMultiplexer 2*1 and 4*1 and 8*1 ➤ Bolean Function Implementation	• Link 1 • Link 2 • Link 3 • Link 4
12.	Basic Flip-Flops (RS.)	Synchronous Sequential Logic	➤ Truth table ➤ Characteristic equation ➤ State diagram ➤ Execution table	• Link 1 • Link 2 • Link 3 • Link 4
13.	clocked RS Flip-Flops			
14.	Analysis of clocked sequential circuits			
15.	JK Flip-Flop			
16.	T Flip-Flop <u>and</u> D Flip-Flop			